

1. The Record Protocol within SSL provides:

- a. transaction's non-repudiation
- b. message confidentiality
- c. message integrity
- d. authentication of entities

2. S/MIME provides protection against:

- a. unauthorized message alteration
- b. e-mail addresses substitution
- c. virus multiplication
- d. denial-of-service attacks

3. Which of the following protocols can be used for the centralized management of users' credentials?

- a. RADIUS
- b. ICMP
- c. ARP
- d. TCACS+
- e. LDAP

4. Which are the main advantages of virtual private networks (VPN) in comparison with frame relay based networks, leased lines or dial-up?

- a. lower installation and operation costs
- b. increasing security level
- c. higher transfer rates
- d. higher availability for network services

5. The "data tunneling" expression, specific to VPNs it refers to:

- a. method used by system administrators for detecting possible network intrusions
- b. a mechanism which allow data to pass through firewall systems
- c. optional feature which activated, increases the network performances
- d. packet encapsulation in other packages belonging to a different protocol in order to create a virtual circuit

6. Which one of the following protocols can be used for creating VPN tunnels?

- a. IPSec
- b. OCSP
- c. PPTP
- d. SSL

7. Which is the SSL protocol used for entities authentication and the negotiation of algorithms and cryptographic keys:

- a. Change Cipher Spec Protocol
- b. Internet Key Exchange
- c. Record Protocol
- d. Handshake Protocol

8. SSL can be used for protecting the following traffic:

- a. ICMP
- b. SMTP
- c. LDAP
- d. FTP

9. Who developed the X.509 standard, describing the digital certificates syntax?

- a. International Telecommunication Union (ITU)
- b. Internet Engineering Task Force (IETF)
- c. International Standards Organization (ISO)
- d. RSA Laboratories

10. How is the session key generated within the Diffie Helman protocol?

- a. the session key is generated by exchanging random values between transmitter and recipient and the use of an algorithm whose security is based on the impossibility of calculating discrete logarithms
- b. the session key is randomly generated by the transmitter and sent encrypted to the recipient
- c. the session key is generated by a third trustful party
- d. the session key is generated simultaneously by transmitter and recipient using a time based algorithm

11. In which of the following authentication protocols mentioned below, the password is transmitted in clear through the network?

- a. Password Authentication Procedure (PAP)
- b. Kerberos
- c. Challenge-Handshake Authentication Protocol (CHAP)
- d. Windows NT LAN Manager (NTLM)

12. Kerberos is efficient against the following:

- a. dictionary attacks
- b. repetition attacks
- c. denial-of service attacks
- d. social engineering

13. Which of the following technologies uses two factors for the authentication of its users?

- a. SAKMP/Oakley
- b. NTLM
- c. LDAP
- d. RSA SecurID

14. The Encapsulating Security Payload (ESP) protocol within IPSec provides:

- a. data origin authentication
- b. data integrity
- c. data confidentiality
- d. data non-repudiation transfer

15. Which of the following sentences about the standard PKCS #7 are true?

- a. it uses the ASN.1 language for describing the data structures
- b. it specifies the syntax for cryptographic messages
- c. it became outdated, being replaced by the PKCS #8 standard
- d. have been developed by the Internet Engineering Task Force (IETF)

16. Which of the following affirmations about the SSL protocol are true?

- a. it uses digital certificates for parties authentication
- b. is a third level protocol
- c. it requires changing the TCP/IP stack
- d. it can only be used for securing the web traffic

17. Which of the following affirmations about the Kerberos protocol are true?

- a. it represents the main authentication mechanism in Windows 2000/2003/2008 domains
- b. has been developed by RSA Security corporation
- c. it uses one time generated passwords
- d. is time dependent

18. Which of the following organizations deals with the standardization of electronic signature at European level?

- a. ETSI ESI
- b. ISO/IEC
- c. PC/SC
- d. CEN/ISSS

19. What are the main advantages of the web of trust model used by PGP to establish trust in a public key:

- a. transitivity of trust relationships
- b. high scalability
- c. doesn't need a third trustful entity
- d. avoids the necessity of key revocation if the key was compromised

20. What needs to be used for encrypting and signing the electronic messages using S/MIME?

- a. S/MIME compatible e-mail servers
- b. S/MIME compatible e-mail clients
- c. SSLization of e-mail servers and clients
- d. digital certificates for users